



MARCO TURRINI

Research Engineer: Scientific Data Analysis & Simulation
Python Pipelines | Machine Learning | Physics-Based Modelling

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marco-turrini-ds

EXPERIENCE

Research Engineer & Project Manager

TNO – Wind Energy

06/2021 – Present

The Netherlands

- Utilised gradient-boosted trees, dense neural networks and Gaussian processes to estimate LiDAR measurements from SCADA data.
- Tested physics-informed neural networks to reconstruct flow-field simulations from LES datasets.
- Applied isolation forests, Hampel filtering and K-means clustering for anomaly detection in wind farm SCADA data.
- Coded and maintained Python data pipelines for multi-sensor measurement data from large-scale offshore sensor networks. Currently developing LLM agent-based tools to support wind farm operational supervision.
- Investigated root causes of discrepancies between field observations and numerical/physical models, combining statistical analysis and domain physics.
- Designed and executed offshore and onshore experimental measurement campaigns: instrumentation planning, field deployment, sensor/LiDAR data acquisition.
- Led collaborative R&D projects (€50k–500k+) and coordinated technical activities between academic, industrial and governmental stakeholders

SUMMARY

Mechanical engineer (EPFL) with 5 years of experience turning large-scale sensor and simulation data into physical insight, in offshore wind energy. Comfortable moving between statistical/ML modelling and physics-based simulation (CFD, wake models, adjoint methods), and building the Python pipelines that connect the two. Motivated by a longstanding interest in physics, CERN, and high-energy particle infrastructure.

Extra Curriculum Activities

Part-time

- Sailing Instructor:** Offshore safety certification (2014 – Present, seasonal)
- Beach Volleyball Coach:** University students (2020 – Present)

EDUCATION

B.Sc & M.Sc. in Mechanical Engineering

EPFL

2014 – 2020

Lausanne, Switzerland

- GPA: 8.8/10.0
- Massachusetts Institute of Technology** Master Thesis: *Settling speed identification of poly-size particulate matter using machine learning* (Grade: 10/10) – experimental sensing, signal processing and ML applied to physical particle behaviour
- Focus on fluid dynamics, heat and mass transport, numerical modeling, sensing techniques

M.Sc Minor in Oceanography

EPFL

2017 – 2020

Lausanne, Switzerland

- GPA: 9.0/10.0
- Offshore field work: 4 weeks on research vessel in Arctic Circle

SKILLS

Data Analysis & Software

Python (Advanced)

NumPy / Pandas / SciPy

Data pipelines

Machine learning

Statistical modelling

Git

Physics & Simulation

Parametric & Monte-Carlo-style studies

Fluid dynamics, Turbulence, Instabilities

Experimental & Sensing

Large-scale sensor networks

Remote sensing (LiDAR)

Signal processing

Engineering Tools

Linux

MATLAB (Advanced)

Dashboards

CAD (Catia V5, Solidworks)

LANGUAGES

French (Native)

Italian (Native)

English (Fluent – C2)

Spanish (Professional – B2)

Dutch (Intermediate – B1)

